ANSWER KEY & MARKING SCHEME SET 03

Full Syllabus (Chapters 1–13) | Class 10 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(b)** yellow	Lead iodide is yellow
2	**(c)** Incorrect — it is neutralised by a base	Acid + base → neutral
3	**(c)** strongly basic	Near 14
4	**(b)** Student A is correct	Pure Cu deposits on cathode
5	**(b)** basic copper carbonate	Moist CO_2
6	**(b)** 3	CH_3COCH_3
7	**(b)** ethyne	$HC{\equiv}CH$
8	**(b)** saprotrophic nutrition	Breakdown outside the body
9	**(c)** heartbeat	Medulla/automatic
10	**(b)** stem cutting	Nodes
11	**(c)** thick wall	Resistant
12	**(d)** XX	Female
13	**(a)** fertilisation	Zygote
14	**(a)** concave	Converges sunlight
15	**(b)** convex	Converging
16	**(a)** Correct	Alloy
17	**(a)** in series	Same current
18	**(d)** in parallel	Same voltage
19	**(c)** interconnected food chains	Network
20	**(d)** DDT	Persists

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	A reaction producing an insoluble solid (precipitate), e.g. $Na_2SO_4 + BaCl_2 \longrightarrow BaSO_4\downarrow + 2\,NaCl$	1+1
22	Salts with the same positive or negative radical, e.g. NaCl and Na_2SO_4 (sodium salts)	1+1
23	Creates an acidic medium for pepsin and kills microbes.	2
24	Muscular tissue and glandular tissue	1+1
25	Seedless plants can be grown; plants mature faster; identical to parent.	2
26	A plant that on self-pollination always produces offspring with the same trait (homozygous).	2
27	Perpendicular distance between the incident ray and the emergent ray in a glass slab.	2
28	Copper has low resistance and is non-magnetic.	2

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	Fe displaces Cu forming green $FeSO_4$. No reaction with Cu in $FeSO_4$, because copper is less reactive than iron.	1+1+1
30	Metals listed in decreasing order of reactivity; Na > Zn > Fe > Cu > Ag	1+2
31	Cells change shape by changing water content; leaves fold and droop quickly without growth	3
32	Body's energy goes to general growth; at puberty: hair growth, oily skin, pimples, genital changes	3
33	Starlight passes through layers of changing refractive index; its apparent position and brightness fluctuate	3
34	Current (directly), number of turns (directly), radius (inversely)	3
35	A — HCl is a strong acid with more H^+ ions than acetic acid (weak acid).	1+2
	OR: Hydrated salts contain water of crystallisation ($CuSO_4 \cdot 5\,H_2O$); anhydrous salts do not ($CuSO_4$).	1+1+1

36	Use cloth/jute bags; reusable bottles and containers; avoid single-use plastic; recycle	3
	OR: Microorganisms have enzymes specific to certain substances; they cannot break down materials like plastic for which no such enzymes exist	3

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	(i) $\frac{1}{v} = \frac{1}{8} - \frac{1}{24} = \frac{1}{12} \Rightarrow v = +12$ cm; $m = v/u = -0.5$; $h' = -1.5$ cm; real, inverted, diminished (ii) magnifying glass/camera	4+1
	OR: (i) 40 cm (2F) (ii) virtual, erect, magnified — no image on screen (iii) $\frac{1}{v} = \frac{1}{20} - \frac{1}{30} = \frac{1}{60} \Rightarrow v = 60$ cm (iv) $m = v/u = -2$ (v) +5 D	1+1+1+1+1
38	(i) soaps: Na/K salts of long-chain carboxylic acids; detergents: ammonium/sulphonate salts of long-chain acids (ii) hydrophobic tails inward in oil, ionic heads outward (iii) their charged ends do not form insoluble precipitates with Ca/Mg ions	1+3+1
	OR: (i) catenation and tetravalency; small size forms strong bonds (ii) CH_4: four shared pairs; C_2H_4: C=C double bond and four C–H (iii) no free ions or electrons	2+2+1
39	(i) $P = VI$, $V = IR \Rightarrow P = I^2R = V^2/R$ (ii) the 100 W bulb; $I = 160/220 = 0.73$ A	2+3
	OR: (i) $(1.5 + 0.1) \times 4 = 6.4$ kWh (ii) ₹44.80 (iii) $R = 220^2/100 = 484\ \Omega$ (iv) heater (v) high resistivity, does not oxidise easily	1+1+1+1+1
40	(i) mouth: chewing, salivary amylase acts on starch; oesophagus: peristalsis; stomach: HCl, pepsin digests protein, mucus protects (ii) acid would damage the stomach lining causing ulcers	4+1
	OR: (i) carries food, O_2, CO_2, wastes, hormones (ii) plasma — transports dissolved substances; RBC — O_2; WBC — defence; platelets — clotting (iii) to push blood around the body under pressure	1+3+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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